



MINING DEALER BEST PRACTICE

Primary Air Filters Removal Tool for 789D OHT's

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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May 2024
1.01-240507-1

1.0 Introduction

Reusing air filters is a good practice recommended by CAT to preserve the environment and save costs. CAT establishes the reuse and cleaning of primary air filters up to two times; this represents a cost savings compared to air filters that were used once and discarded due to the extraction process.

In this operation, some complications have occurred when removing the primary air filters on the 789D as they get stuck inside the pre-cleaner when the machines are in PM (Preventive Maintenance). In some situations, the filters become so stuck that the technicians have to drill into the filters with a pry bar and attempt to pull them out. Once the filter is punctured, it can no longer be reused, and costs will increase due to the need to acquire more new filters to replace the discarded ones.



This Best Practice is intended to make a tool to easily remove the air filters from the 789D without damaging them.

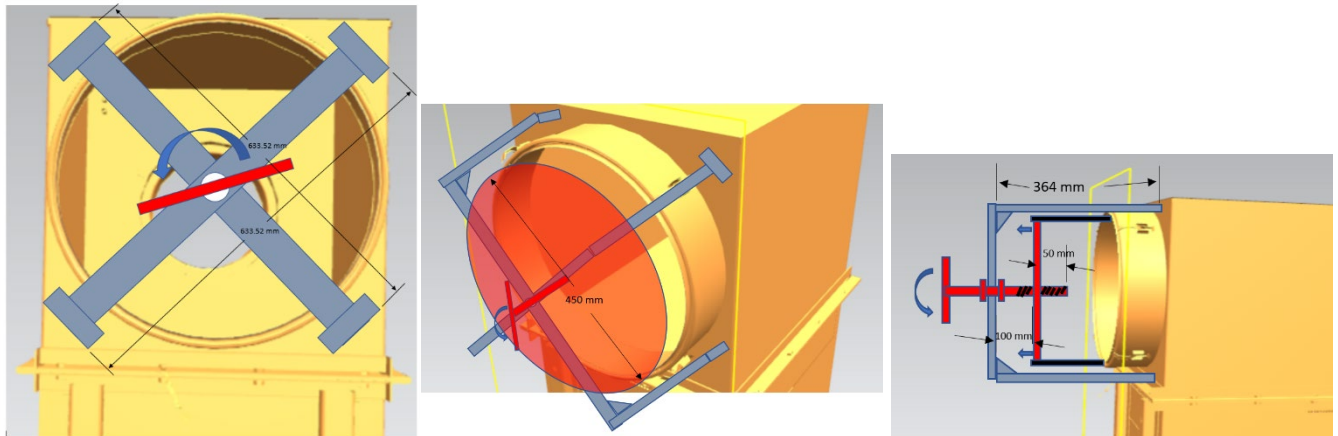
2.0 Best Practice Description

This Best Practice intends to use a removal tool to easily remove the primary air filters during PM in the 789D. The tool must be attached to the filter to be removed and supported to the pre-cleaner. Once the tool is attached to the filter, the technician must turn a handle a few times and the filter will come out of the pre-cleaner. It is not necessary to remove the entire filter from the pre-cleaner, removing the filter one inch will be enough.

The tool must be removed in the opposite way of installation. The filter can now be easily removed by hand from the pre-cleaner.

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The tool is manufactured with a bracket with 4 supports and a worm screw welded to a rotating handle. The tool also incorporates a round plate, a hard washer, a nut, and a lock nut. There is a rubber band supported to the round plate with a clamp. When removing the air filter, tighten another clamp on the opposite side of the rubber band to tighten the filter.



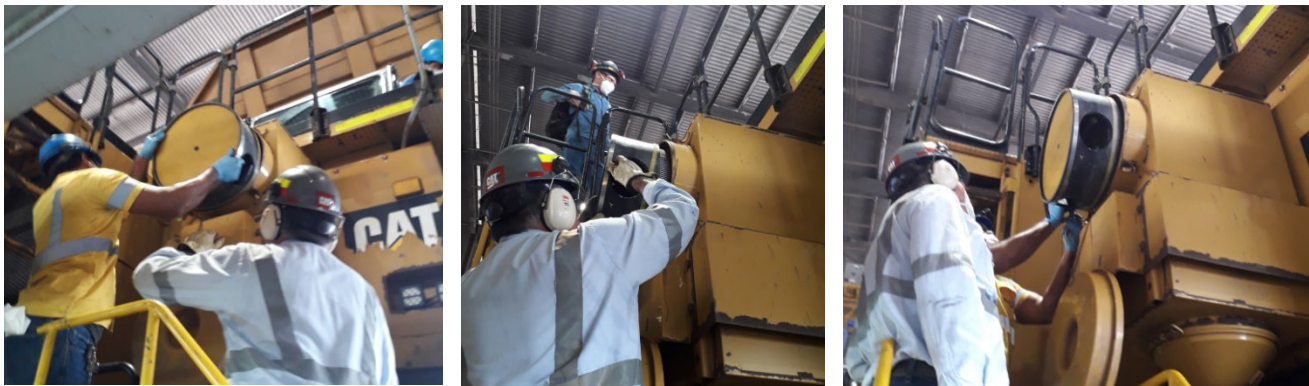
3.0 Implementation Steps

A. Remove the pre-cleaner cover.



B. Attach the plate with the rubber band to the filter and tighten the clamp to it.

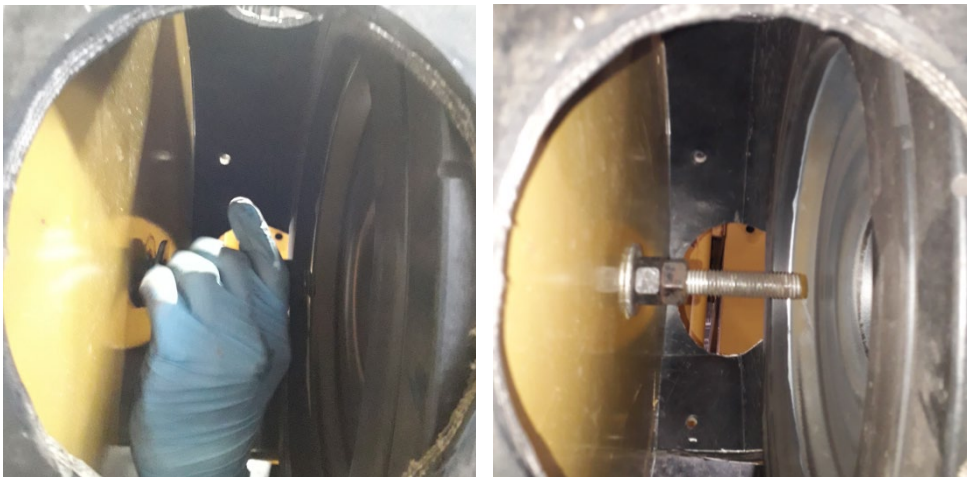
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C. Attach the bracket to the plate by inserting the tool pin into the hole in the plate.



D. Attach the bolt with a hard washer, locknut, and nut to the plate. Tighten the nut with the lock nut to prevent them from turning freely relative to the bolt when removing the filter.



E. Turn the tool handle counterclockwise and you will begin to see the filter come out of the pre-cleaner. Continue turning until the filter comes out approximately one inch.

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- F. Remove the tool in the reverse order of installation.
- G. Finish removing the filter from the pre-cleaner by hand.
- H. Perform the same procedure for the air filter on the opposite side.

4.0 Benefits

This tool will facilitate the removal of the primary air filters on the 789D. These filters are prone to get stuck inside the pre-cleaner when trying to remove them in the PM. In some cases, the filter must be punctured with a pry bar to remove it and can no longer be reused. The tool facilitates the removal of the filters and maintains their functionality. The filter can be reused two more times after the first use.

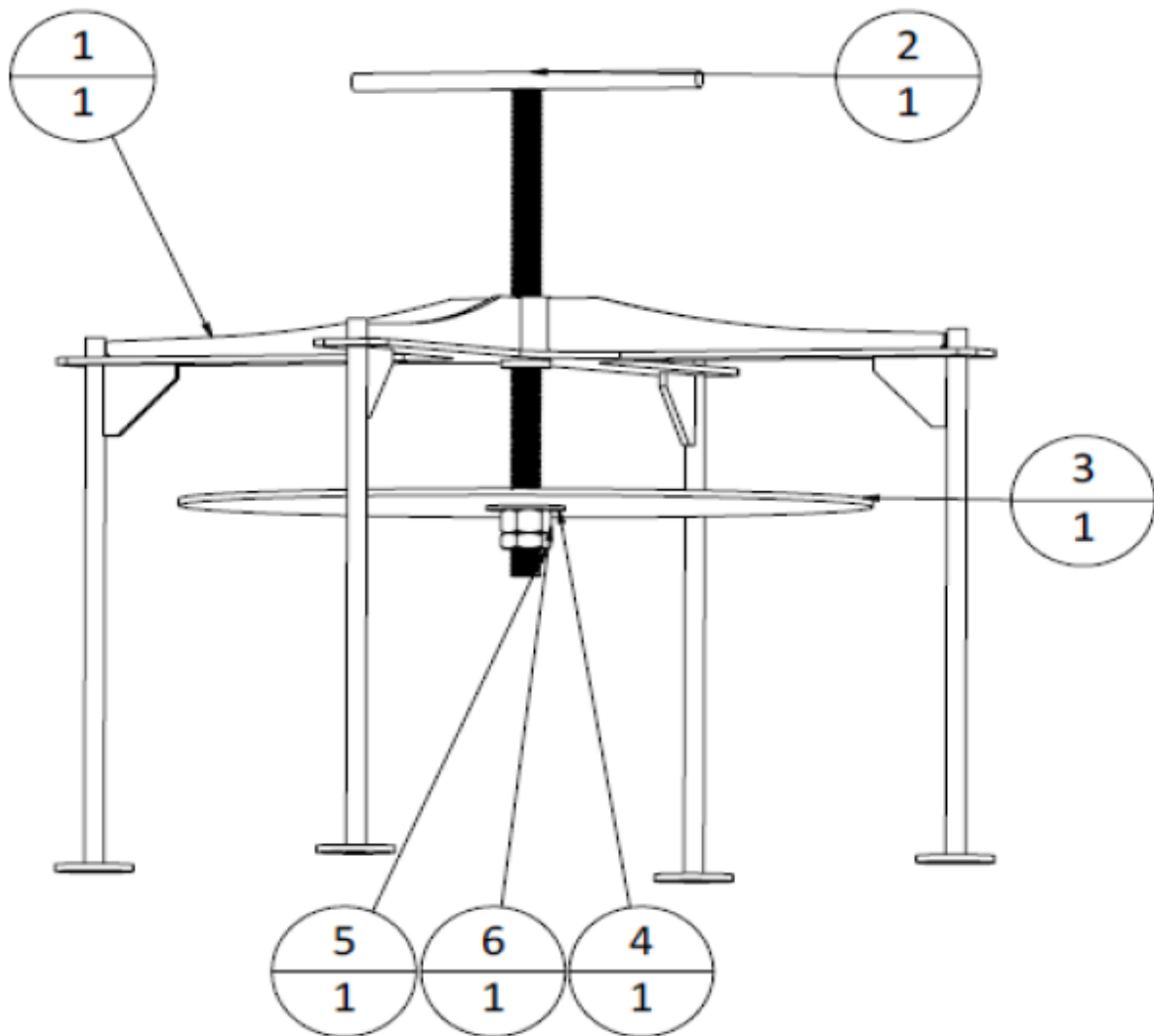
5.0 Resources Required

Investment

Parts and fabrication materials = US\$ 600.

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6.0 Supporting Attachments / References



Item	Description	QTY
1	Bracket	1
2	Turning Handle	1
3	Plate	1
4	Washer	1
5	Nut	1
6	Locknut	1
---	Rubber band	1
---	Clamps	2

- All relevant drawings are included in the appendix.

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7.0 Related Best Practices

None

8.0 Acknowledgements

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